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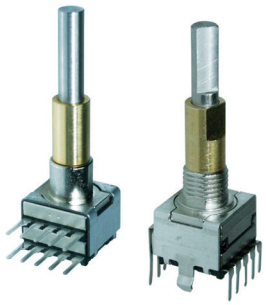
# Product description

## MAIN FEATURES

DUAL FUNCTION AND HIGH PERFORMANCE

- Dual encoder: Two encoders - one space
- Resolution: 16 or 32 detent
- With or without integrated push button
- Rotational life: Up to 1 Million revolutions
- Excellent indexing feel with 0.5, 1, 1.5, 2, 2.5, 3, 3.5 or 4.5 Ncm switching torque (remains consistent over life)
- Gold plated contacts
- Robust metal housing
- Body size: 11.5 x 12.3 x 9.1 mm
- IP68 shaft and front panel sealing
- Operating temperature: -40 to +85 °C
- Shaft electrically insulated > 500 VDC (Shaft to contact system)
- Various options and customizations

**E37**



Pb ELV (2000/53/EC)
RoHS (2011/65/EU)

MIL-STD-202G

SWISS CLICK INDEXING SYSTEM™
(for more information see chapter «Technical explanations»)

## PRODUCT VARIETY

- Vertical or horizontal mounting
- Threaded or non-threaded bushing
- Push button force 3, 6, 10, 14 N or without push
- Resolution / pulses per revolution (PPR) 32 / 16, 32 / 8, 16 / 16, 16 / 8
- Switching torque 0.5, 1, 1.5, 2, 2.5, 3, 3.5 or 4.5 Ncm or no detent
- Front panel sealing IP60 or IP68

## POSSIBLE CUSTOMIZATIONS

- Shaft dimension and shape
- Stainless steel housing
- Switching torque and push button actuation force
- Indexing resolution and PPR

## TYPICAL APPLICATIONS

- Cockpit control, radios and navigation
- Desktop and mobile radios
- Professional, portable audio equipment
- Applications where user interface is space critical

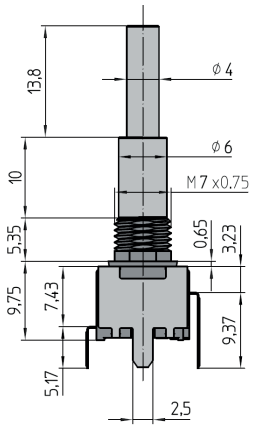
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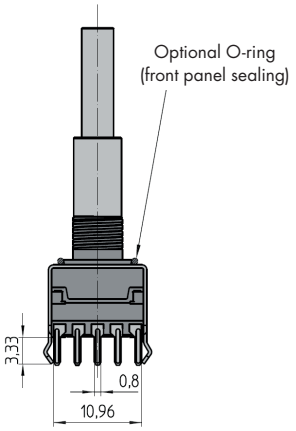
# Dimensions and pin assignment

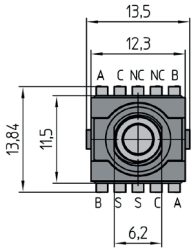
## SWITCH DESIGN

### THT VERTICAL

Example of illustration with thread



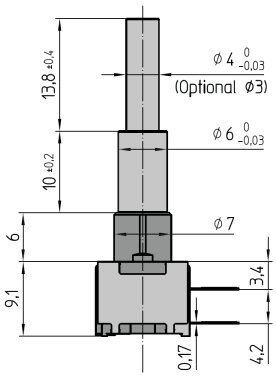



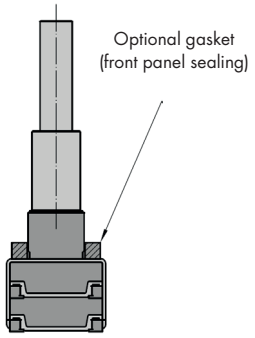



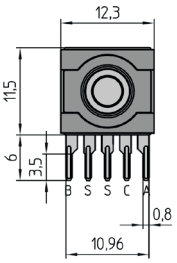

Both threaded and non-thraded bushing are available for all versions; THT vertical or THT horizontal (see type key).

### THT HORIZONTAL

Example of illustration without thread

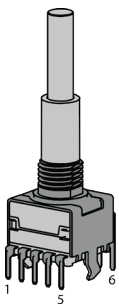


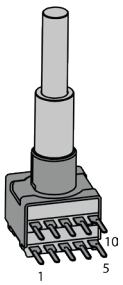




Both threaded and non-thraded bushing are available for all versions; THT vertical or THT horizontal (see type key).

## PIN ASSIGNMENT




vertical




horizontal

| PIN | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | Bi | S | S | Ci | Ai |
| | 6 | 7 | 8 | 9 | 10 |
| | Bo | NC | NC | Co | Ao |

Dimensions in mm
Tolerances according to DIN ISO 2768-1 (m), unless otherwise specified

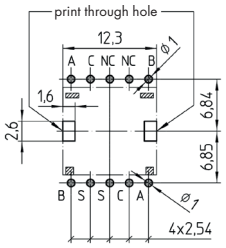
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# Dimensions and pin assignment

## DRILLING AND FOOTPRINT

THT VERTICAL

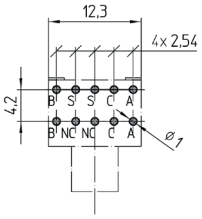
View from component side of the PCB




PCB-thickness: 1 to 1.5 mm

THT HORIZONTAL

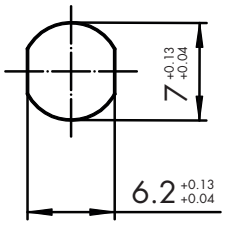
View from component side of the PCB



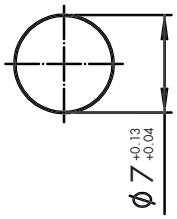

PCB-thickness: 1 to 1.5 mm

## FRONT PANEL CUT OUT

THREADED

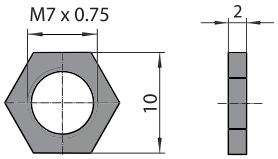



NON-THREADED



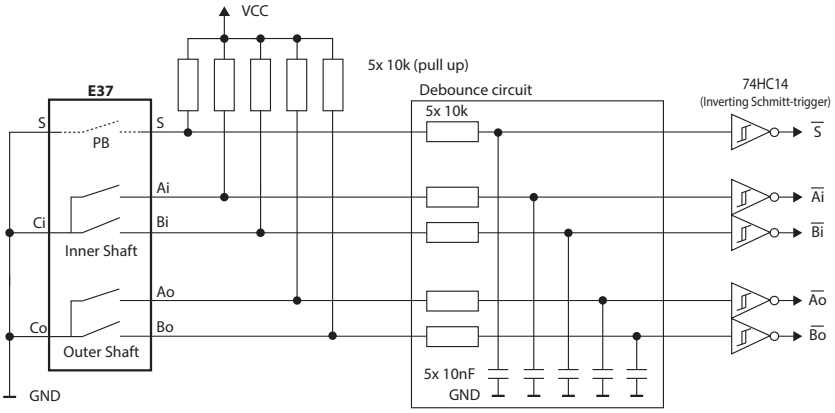

## NUT

HEX NUT (SUPPLIED)




# Circuit diagram

## RECOMMENDED SYSTEM INTERFACE



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# Ordering information

## ORDERING CODE

| E37 | - | _ | _ | _ | _ | _ | _ | - | _ |
|---|---|---|---|---|---|---|---|---|---|

### ORIENTATION | MOUNTING

**V** THT vertical
**C** THT horizontal

### BUSHING

**T** Threaded M7 x 0.75 x 6 mm
(nut supplied, packed separately)
**N** Non-threaded Ø 7 x 6 mm

### PUSH BUTTON

**0** Without push button
**3** 3 N
**6** 6 N
**A** 10 N
**E** 14 N

### INNER SHAFT[1] | RESOLUTION | TORQUE

| | | | |
|---|---|---|---|
| **2** | 16 detent | (8 PPR) | 1.5 Ncm |
| **3** | 16 detent | (8 PPR) | 2.5 Ncm |
| **4** | 16 detent | (16 PPR) | 1.5 Ncm |
| **6** | 16 detent | (16 PPR) | 2.5 Ncm |
| **8** | No detent | (16 PPR) | |
| **9** | No detent | (8 PPR) | |
| **G** | 16 detent | (8 PPR) | 4.5 Ncm |

[1] O-ring of IP65 / IP68 shaft sealing may slightly increase switching torque

### OUTER SHAFT[1] | RESOLUTION | TORQUE

| | | | |
|---|---|---|---|
| **1** | 32 detent | (16 PPR) | 2 Ncm |
| **2** | 16 detent | (8 PPR) | 1.5 Ncm |
| **3** | 16 detent | (8 PPR) | 2.5 Ncm |
| **4** | 32 detent | (8 PPR) | 2 Ncm |
| **5** | 16 detent | (16 PPR) | 1.5 Ncm |
| **6** | 16 detent | (16 PPR) | 2.5 Ncm |
| **8** | No detent | (16 PPR) | |
| **9** | No detent | (8 PPR) | |
| **A** | 32 detent | (16 PPR) | 0.5 Ncm |
| **B**[2] | 32 detent | (16 PPR) | 1 Ncm |
| **C** | 32 detent | (16 PPR) | 1.5 Ncm |
| **D** | 32 detent | (16 PPR) | 3 Ncm |
| **E** | 16 detent | (8 PPR) | 0.5 Ncm |
| **F** | 16 detent | (8 PPR) | 3.5 Ncm |
| **G** | 16 detent | (8 PPR) | 4.5 Ncm |

[1] O-ring of IP65 / IP68 shaft sealing may slightly increase switching torque
[2] Available with non-threaded bushing only

### IP SEALING

**0** IP60
**1**[1] IP68 shaft sealing
**2**[1] IP68 shaft | front panel sealing
(non-threaded bushing gasket provides IP65,
O-ring | gasket is mounted)

[1] O-ring of IP65 / IP68 shaft sealing may slightly increase switching torque

### SHAFT TYPE

**1** Basic (see drawing)
**2** Inner shaft; Ø 3 mm, same length
as basic type

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# Ordering information

## PREFERENCE TYPES SELECTION CHART[1]

| PUSH BUTTON | INNER SHAFT | OUTER SHAFT | IP SEALING | PART NUMBER THT VERTICAL (THREADED BUSHING) | PART NUMBER THT HORIZONTAL (THREADED BUSHING) |
|---|---|---|---|---|---|
| Yes, 6 N | 16 detent (8 PPR) 2.5 Ncm | 16 detent (8 PPR) 2.5 Ncm | IP60 | E37-VT6330-1 | E37-CT6330-1 |
| | | | IP68 | E37-VT6332-1 | E37-CT6332-1 |
| | | 32 detent (16 PPR) 2 Ncm | IP60 | E37-VT6310-1 | E37-CT6310-1 |
| | | | IP68 | E37-VT6312-1 | E37-CT6312-1 |

## PACKAGING

Blister box: 20 pieces (nuts are supplied and packed separately)

## ACCESSORIES AND SPARE PARTS

Hex nut M7 X 0.75: Part number 4516-40 (50 pieces / bag), brass, nickel plated

RECOMMENDED KNOBS



| CLASSIC COLLETS 10 / 14.5 MM (FOR SHAFT TYPE Ø 3 MM) | | | |
|---|---|---|---|
| Inner shaft | Cap | 10 mm black, glossy | 040-1020 |
| | Knob | 10 mm, classic collet, glossy | 020-2120 |
| Outer shaft | Knob | 14.5 mm, classic collet, glossy | 020-3440 |
| Inner shaft | Cap | 10 mm black, matt | 040-1025 |
| | Knob | 10 mm, classic collet, matt | 020-2125 |
| Outer shaft | Knob | 14.5 mm, classic collet, matt | 020-3445 |



| METAL KNOBS 11 / 15 MM (FOR SHAFT TYPES Ø 4 MM) | |
|---|---|
| Inner shaft | 11 mm, metal |
| Outer shaft | 15 mm, metal |
| **2 pc sets:** | |
| Silver: CAE041559 | |
| Black: CAE041560 | |

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[1] For other types | options see ordering code

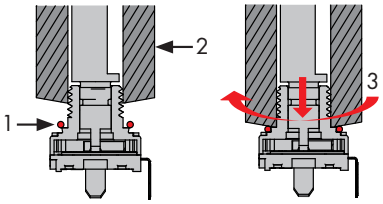
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# Ordering information

## ACCESSORIES AND SPARE PARTS



Part number: E33-ORING-TOOL




1 Slip the lubricated O-ring over the bushing.
2 Slide the mounting tool over the bushing.
3 While pushing down the O-ring simultaneously rotate the mounting tool.

# Specifications

### Mechanical data

| | |
|---|---|
| Positions: | Inner shaft: 16 positions or no detent<br>Outer shaft: 16, 32 positions or no detent |
| Switching torque: | Inner shaft: 16 positions with 1.5, 2.5 Ncm, 4.5 Ncm or without detent (±30 % in new condition)<br>Outer shaft: 32 positions with 0.5, 1, 1.5, 2 or 3 Ncm (±30 % in new condition)<br>16 positions with 0.5, 1.5, 2.5, 3.5, 4.5 Ncm or no detent (±30 % in new condition) |
| Rotational life: | > 1'000'000 revolutions with 0.5, 1 or 1.5 Ncm switching torque or no detent<br>> 500'000 revolutions with 2 Ncm switching torque<br>> 300'000 revolutions with 2.5 Ncm switching torque<br>> 100'000 revolutions with 3, 3.5 or 4.5 Ncm switching torque<br>(tested at room temperature) |
| Allowed shaft load: | 50 N push, 50 N pull and 50 N side load (static at 20 mm from support surface) |
| Fastening torque of nut (front panel mounting): | M7 x 0.75: < 100 Ncm |

### Electrical data

| | |
|---|---|
| Electrical connection: | Pins 0.23 x 0.8 mm |
| Switching voltage: | < 15 VDC (resistive load) |
| Switching current: | < 10 mA (resistive load) |
| Contact resistance: | < 10 Ω (over the entire rotational life) |
| Signal \| coding: | 2-Bit quadrature |
| Resolution (pulses per revolution): | 16 or 8 PPR per channel |
| Rotational speed: | < 60 rpm |
| Phase shift: | 90° (±70°) |
| Contact bouncing: | < 2 ms (at 60 rpm) |
| Dielectric strength: | 500 VDC during 60 s (MIL-STD-202G, method 301) |
| Insulation resistance: | > 1 GΩ at 500 VDC (in new condition) |

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# Specifications

## MATERIALS

| | |
|---|---|
| Shaft: | Inner shaft: Stainless steel 1.4305<br>Outer shaft: Brass CuZn38Pb2 |
| Bushing \| housing: | Zinc die casting (nickel plated), fiberglass reinforced high performance plastic |
| Contact surface: | Cu alloy (Au plated) |
| Soldering leads: | Cu alloy (tin plated) |
| Hex nut: | Brass (nickel plated) |
| Housing clamp: | Tinplate |
| O-rings: | NBR (nitrile rubber), 70 shore A |
| Front panel sealing: | Threaded bushing: O-ring<br>Non-threaded bushing: EPDM-rubber, 45 shore A, complies with SAE J 18-79 |

## ENVIRONMENTAL DATA

| | |
|---|---|
| Operating temperature: | -40 to +85 °C (IEC 60068-2-14) |
| Storage temperature: | -65 to +125 °C (IEC 60068-2-14, MIL-STD202G, method 107G, condition B-3) |
| Humidity: | < 93 % relative humidity (MIL-STD-202G, method 103B, condition B) |
| IP sealing against front panel: | IP60 without sealing<br>IP65 with non-threaded bushing, shaft and front panel sealing<br>IP68 with threaded bushing, shaft and front panel sealing (2 bar, 1 h) |
| Vibration: | 29 $G_{RMS}$ at 100 to 1'000 Hz (MIL-STD-202G, method 214A, condition 1 h / 15 min) |
| Shock: | 100 G (MIL-STD-202G, method 213B, condition C) |
| Flammability: | UL94-V0<br>Gaskets UL94-HB |

## SOLDERING CONDITIONS

| | |
|---|---|
| Hand soldering: | < 300 °C during 3 s |
| Reflow soldering: | < 280 °C during 5 s |

## MECHANICAL DATA FOR PUSH BUTTON

| | |
|---|---|
| Actuation force: | 3, 6, 10 or 14 N (±30 % in new condition) |
| Travel: | 0.5 (±0.2) mm |
| Lifecycles: | > 200'000 cycles (tested at room temperature) |

## ELECTRICAL DATA FOR PUSH BUTTON

| | |
|---|---|
| Switching voltage: | < 15 VDC (resistive load) |
| Switching current: | < 10 mA (resistive load) |
| Contact bouncing: | < 2 ms (at 2 Hz) |

## MATERIALS FOR PUSH BUTTON

| | |
|---|---|
| Contact surface: | Cu alloy (Au plated) |
| Snap dome: | Stainless steel (Au plated) |

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